Service Bulletin

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Biodiesel Blend Operational Guidelines for Indonesia	

Biodiesel Blend Operational Guidelines for Indonesia

Purpose:

As of July 1, 2026 the Indonesian government has mandated that biodiesel should be blended in diesel fuel at a 50 percent rate for certain fuel grades which are sold within the country. Since the maximum biodiesel blending rate for many Cummins engines is 20 percent, some guidance is required when using higher blends. This document defines the conditions under which biodiesel blend levels that exceed the limits specified in Fluids for Cummins® Products Service Manual, Bulletin 5411406 may be used in Cummins® engines.

Scope:

This Service Bulletin applies **only** to non-road engines in Indonesia without exhaust aftertreatment.

Required Operational Conditions:

The following categories detail criteria which, when followed, can mitigate risk when operating with biodiesel blends that exceed Cummins established guidelines. Failure to follow these guidelines can result in equipment downtime and costly repairs.

- Fuel Quality Requirements
- Biodiesel Guidelines for Operation
- Fuel Filtration Recommendations
- Elastomer Impacts
- Lubricating Oil Drain Guidance
- Fuel Storage and Handling Guidelines

1. Fuel Quality Requirements

B100 blended to 50 percent in conventional diesel should conform to the follow requirements as per Indonesian Ministry of Energy and Mineral Resources Number **257.K/EK.01/MEM.E/2026.**

Fuel properties from this specification are listed in Table 1 for convenience.

Table 1, Fuel properties

Parameter	Units	Minimum	Maximum	Test Method
Density 15°C [59°F]	kg/m3	850	890	ASTM D4052
Viscosity 40°C [104°F]	mm2/s (cSt)	2.3	6	ASTM D445
Cetane Number		51		ASTM D613
Flash Point (Closed Cup)	0°C [32°F]	130		ASTM D93
Copper Corrosion (3 hours @ 50C)		Number 1		ASTM D130
Ramsbottom Carbon Residue (10% distillation Residue)	mass %		0.3	ASTM D4530
Distillation Temperature at 90% recovered	0°C [32°F]		360	ASTM D1160
Sulfated Ash	mass %		0.02	ASTM D874
Sulfur content	mg/kg		10	ASTM D5453
Phosphorous Content	mg/kg		4	SNI 7182:2015 AOCS Ca 12-55
Acid Number	mg KOH / g		0.4	ASTM D664
Free Glycerin	mass %		0.02	ASTM D6584
Total Glycerin	mass %		0.24	ASTM D6584
Iodine Number	mass %		115	SNI 7182:2015 AOCS CD 1-25
Methyl Ester Content	mass %	96.5		SNI 7182:2015
				ASTM D8274
Oxidation Stability	Minutes	900		EN 15751
Rapid Small Scale Oxidation Test (RSSOT)	Minutes mass %	67.5		ASTM D7545
Monoglyceride content			0.47	ASTM D6584
Color			3	ASTM D1500
Water Content	mg/kg		300	ASTM D6304

Table 1, Fuel properties				
Parameter	**Units**	**Minimum**	**Maximum**	**Test Method**
Cold Filter Plugging Point (CFPP)	0°C [32°F]		15	ASTM D6371
Group I Metals (Na+K)	mg/kg		5	ASTM D7111
Group II Metals (Ca+Mg)	mg/kg		5	ASTM D7111
Total Contamination	mg/liter		20	ASTM D6217

2. Biodiesel Guidelines for Operation

To successfully use biodiesel blends greater than 20 percent, it is imperative that the fuel is of high quality. Users of Cummins® engines are encouraged to confirm with fuel suppliers that the biodiesel used to blend with diesel in Indonesia conforms to the requirements of the fuel quality specification listed in Table 1. In addition, engine filtration guidelines below should be followed. See section 3.

3. Recommended Fuel Filtration

To successfully use blends greater than B20, fuel filtration **must** be enhanced beyond the level required for standard diesel fuel. Biodiesel has an affinity for water which can accelerate fuel degradation and promote microbial growth. Field experience with biodiesel blends greater than B20 indicates an increased risk of fuel filter plugging. Therefore, a conservative maintenance approach is recommended, with fuel filters replaced at 50 percent of the normal service interval.

Note: Following successful field validation, the filter replacement interval may be increased. Additionally, the filter seal material could affect change interval. Check with OEM for recommendations.

The primary sources of filter blockage include oxidation and aging by-products, monoglyceride and wax precipitation, and metal soaps and other contaminants. Fuel filtration should specifically be used to mitigate biodiesel characteristics and that systems include multi-stage filtration architecture, advanced water-separation, higher contaminant-holding capacity, and fully synthetic filter media. Table 2 shows the minimum requirements for each filtration stage:

Table 2, Minimum requirements for each filtration stage			
Common Specifications	**Stage 0**	**Stage 1**	**Stage 2**
Media Type	Synthetic	Synthetic	No Changes
Water Removal (SAE J1488)	0.95	0.95	
Water in Fuel Sensor (WIF)*		Y	
Maximum fuel supply restriction at fuel pump inlet with clean fuel filter element(s) at maximum fuel flow	Per engine family datasheet		
F3.8/B4.5/B6.7/L9 Specifications			
Particle Removal (ISO 19438) (minimum)	≥99.7%@10μm	≥99.0%@5μm	
Total Water Capacity (minimum)	1,050 ml		
Total Particle Capacity (minimum)	360g		
X12/X15 Specifications			
Particle Removal (ISO 19438) (minimum)	≥98.7%@10μm ≥85%@4μm	≥98.7%@5μm ≥90%@4μm	
Total Water Capacity (minimum)	1,050 ml		
Total Particle Capacity (minimum)	535g		

*Recommend setting up WIF sensor driven engine derate when possible.

4. Elastomer impacts

Most Cummins® engines are designed with standard Fluoropolymer o-ring, and seal materials. These are present in fuel wetted elastomer components and are robust for greater than B20 blends. Stage 0 and/or Stage 1 filtration options commonly purchased with the engine as an FS option, were **not** designed with elastomers for usage above B20 blends. They also were sized and validated for up to B20 and typical fuel quality experienced in the field. Therefore, they should **not** be used for greater than B20 blends or in cases where fuel conditions are consistently at the upper end of water content or particulate matter allowances as customers will experience shorter than desirable water draining and filter change intervals. In the case where the FS option is supplied with hoses, these were also **not** designed or validated for usage in biodiesel blends greater than B20 and should **not** be used in these applications.

5. Lubricating Oil Drain Guidance

Lubricating oil drain interval (ODI) is based on each engine platform. Follow guidelines from engine owner's manuals and Fluids for Cummins® Products Service Manual, Bulletin 5411406 with the exception that lubricating oil drain intervals should be reduced by 50 percent for engines which run greater than B20 fuel blends. Through successful used lubricating oil testing and validation, lubricating oil drain intervals may be increased. Fuel dilution of lubricating oil has been observed with the operation of biodiesel under certain operating conditions. Fuel dilution monitoring can be accomplished by performing oil sampling and levels in lubricating oil **must not** exceed 5 percent. Additional information on lubricating oil contamination and oil sampling can be found in Cummins® Engine Oil and Oil Analysis Recommendations, Bulletins 3810340, and Fluids for Cummins® Products Service Manual, Bulletin 5411406.

6. Storage & Handling

Biodiesel is very sensitive to storage conditions and has a short shelf life. If **not** properly maintained, it may affect the engine performance and could lead to hardware replacement. Some common problems associated with poor quality biodiesel are fuel system deposits and corrosion, plugged/ shortened filter life, poor cold weather performance, and microbial growth. Other inherent properties of biodiesel may cause customers to experience operational issues such as poor on-engine fuel – water separation, decreased fuel economy and/or power. It is recommended that biodiesel fuel be consumed within 3 months of purchase/delivery. Further information can be found in section 375-004 Biodiesel in Fluids for Cummins® Products Service Manual, Bulletin 5411406

Warranty Statement

This letter does **not** supersede any of the commercial terms of sale, including, but not limited to, warranty coverage.

Document History

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2026-8-13	Module Created